



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to K.N.S. AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

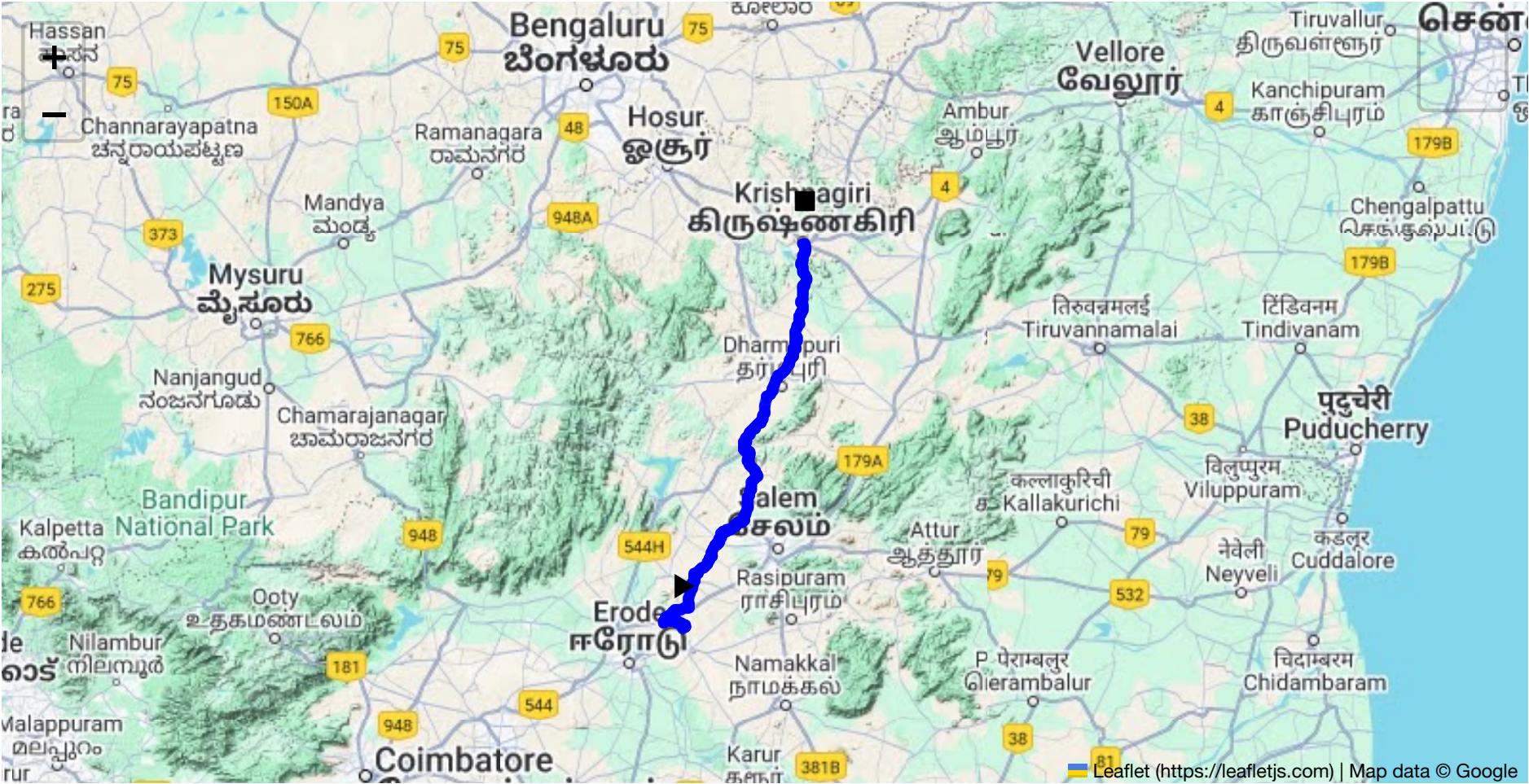
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 151.38 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours

Start: (11.4381, 77.8734)
End: (12.522651, 78.219139)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 151.38 kilometers, originating from Sangagiri and ending at Krishnagiri, passing through several key points, including Bhavani Main Road, SH 86, and Salem. The path mainly traverses state highways and major roads that are frequented by heavy vehicles.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with the following expected conditions:

- **Summer (March to June):** High temperatures, potential heating hazards affecting vehicle performance.
- **Monsoon (July to September):** Heavy rainfall can result in reduced visibility, slippery roads, and minor flooding in low-lying areas.
- **Post-Monsoon and Winter (October to February):** Generally mild conditions, but occasional fog may reduce visibility in early mornings.

3. Analysis of Traffic Patterns

- **Peak Hours:**
 - Morning: 8 AM to 10 AM
 - Evening: 5 PM to 8 PM
- **Congestion-Prone Areas:**
 - **Salem City:** High traffic due to urban density, particularly on main roads.
 - **Bhavani Main Road and SH 86:** Busy during agricultural seasons due to local market freight.

4. Assessment of Road Quality and Infrastructure

- **General Road Quality:** Most segments are well-maintained state highways; however, minor roads in and around Salem may experience wear and tear.
- **Infrastructure:** Adequate signage and road markings, occasional construction in expanding urban areas can cause temporary hazards.

5. Suggestions for Alternative Routes for Emergencies

- Utilize NH 44 to bypass Salem if congested.
- NH 544 serves as a viable alternate parallel route providing bypass options to Krishnagiri.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Timings:** Restrictions may apply in specific urban areas during peak hours.
- **Mandatory Safety Measures:** Inspection compliance check, placards on vehicles, and specific permits required for transporting hazardous materials.

7. Overview of Historical Incidents

No significant historical incidents involving heavy vehicles or hazardous materials have been recorded on this route specifically, but adjacent areas have seen minor incidents due to slippery roads during monsoon season.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:**
 - Agricultural zones near Bhavani Main Road.
 - Waterways around Salem may require caution to prevent contamination.
- **Environmental Concerns:** Minimize emissions around populated areas, and ensure no spillage in agricultural lands.

9. Analysis of Communication Coverage

- **Coverage:** Generally reliable in urban and semi-urban sections.
- **Dead Zones:** Rural stretches, primarily between Salem and Kaveripattinam, may have weak signals intermittently.

10. Estimated Emergency Response Times

- **Urban Segments (Salem):** Approximately 15-20 minutes.
- **Rural Areas:** Up to 30-45 minutes due to sparse facilities and potential access issues.

12. Overall Summary of Risk Assessment

The route is largely safe for heavy vehicle transport with standard preparations for weather and traffic conditions. Significant care should be taken during peak traffic hours and adverse weather conditions. Alternative routes should be planned ahead of time to ensure smooth navigation in case of unforeseen

circumstances. Monitoring local regulations and ensuring compliance with safety standards are essential to minimize risks associated with hazardous material transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	Medium	11.45546, 77.81800	30 KM/Hr
7	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.56072, 77.89919	30 KM/Hr
11	Turn	Medium	11.56140, 77.89991	30 KM/Hr
12	Turn	Medium	11.59085, 77.90592	30 KM/Hr
13	Turn	Medium	11.59253, 77.90690	30 KM/Hr
14	Turn	Medium	11.63848, 77.94825	30 KM/Hr
15	Turn	High	11.64197, 77.95197	15 KM/Hr
16	Turn	Medium	11.64558, 77.95108	30 KM/Hr
17	Turn	Medium	11.67883, 77.96865	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71386, 77.99552	30 KM/Hr
20	Turn	Medium	11.71735, 78.00019	30 KM/Hr
21	Turn	Medium	11.72765, 78.02291	30 KM/Hr
22	Turn	High	11.73785, 78.03496	15 KM/Hr
23	Turn	High	11.73653, 78.03886	15 KM/Hr
24	Turn	Medium	11.74252, 78.04705	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	Medium	11.87998, 78.08170	30 KM/Hr
28	Turn	Medium	11.91266, 78.05852	30 KM/Hr
29	Turn	Medium	11.92653, 78.06517	30 KM/Hr
30	Turn	Medium	11.93746, 78.05331	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	11.95531, 78.04698	30 KM/Hr
32	Turn	High	11.95854, 78.04990	15 KM/Hr
33	Turn	High	11.96015, 78.04945	15 KM/Hr
34	Turn	Medium	11.96370, 78.05142	30 KM/Hr
35	Turn	Medium	11.96433, 78.05113	30 KM/Hr
36	Turn	High	11.96542, 78.04938	15 KM/Hr
37	Turn	High	11.96650, 78.04927	15 KM/Hr
38	Turn	Medium	12.17041, 78.16823	30 KM/Hr
39	Turn	Medium	12.22515, 78.19158	30 KM/Hr
40	Turn	Medium	12.22995, 78.19466	30 KM/Hr
41	Turn	Medium	12.24547, 78.19501	30 KM/Hr
42	Turn	Medium	12.24685, 78.19561	30 KM/Hr
43	Turn	Medium	12.25120, 78.20010	30 KM/Hr
44	Turn	Medium	12.25442, 78.20015	30 KM/Hr
45	Turn	Medium	12.32012, 78.20734	30 KM/Hr
46	Turn	Medium	12.40661, 78.22318	30 KM/Hr
47	Turn	Medium	12.42548, 78.22666	30 KM/Hr
48	Turn	Medium	12.45066, 78.21821	30 KM/Hr
49	Turn	Medium	12.50265, 78.22345	30 KM/Hr
50	Turn	High	12.50736, 78.22244	15 KM/Hr
51	Turn	High	12.50803, 78.22286	15 KM/Hr
52	Turn	High	12.51930, 78.22187	15 KM/Hr
53	Turn	High	12.52231, 78.21904	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
34	hospital	TCR Multi Speciality Hospital	12.5088183,	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
			78.2237657		
36	hospital	Sri Saraswathi Hospital- Ortho Hospital	12.51332, 78.221676	30 km/h	Medium
37	hospital	Dr. Agarwals Eye Hospital Ltd., Krishnagiri	12.5202697, 78.2189856	30 km/h	Medium
38	hospital	Ganesh Maternity Hospital	12.520064, 78.2198424	30 km/h	Medium
39	hospital	Sristi Children's Hospital	12.519915, 78.219104	30 km/h	Medium
40	hospital	Sabari Eye Hospital	12.5190949, 78.2190115	30 km/h	Medium
41	hospital	Kalyani ENT Hospital	12.5202161, 78.2191611	30 km/h	Medium
42	clinic	Rakshana Hearing Aid Centre	12.519233, 78.218876	30 km/h	Medium
43	hospital	Sre-Vidya Hospital	12.518789, 78.2186571	30 km/h	Medium
44	clinic	Padma Poly Clinic	12.5224009, 78.2191314	30 km/h	Medium
45	hospital	Ramacharan ENT Hospital	12.5192713, 78.2187027	30 km/h	Medium
46	hospital	Kaveri Children Hospital	12.5191998, 78.2185048	30 km/h	Medium
47	hospital	Maharishi Siddha Hospital	12.5192281, 78.2186874	30 km/h	Medium
48	hospital	Padma Pharmacy	12.5227462, 78.2191806	30 km/h	Medium
51	hospital	Government Hospital	12.5246441, 78.2199165	30 km/h	Medium
52	hospital	Jeeva Hospital	12.523671, 78.2175399	30 km/h	Medium
53	hospital	Nathans Speciality hospital	12.5203307, 78.2174264	30 km/h	Medium
54	clinic	A R K Nursing Home	12.5207422, 78.2166169	30 km/h	Medium
55	hospital	Sri Sathiya Sai Hospital	12.523345, 78.2171131	30 km/h	Medium
56	clinic	Chennai Diabetic Centre	12.520646, 78.2168832	30 km/h	Medium
57	clinic	KMS Child Care	12.520149, 78.2182649	30 km/h	Medium
60	police	KRishnagiri Police Station	12.5253089, 78.2168454	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
32	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
33	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
35	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
49	school	Government Boys higher secondary school	12.5233938, 78.2200134	30 km/h	Medium
50	school	Government Girls Higher Secondary School	12.5240408, 78.21795	30 km/h	Medium
58	school	Govt Boys High Sec School	12.523304, 78.2207964	30 km/h	Medium
59	school	Government Girls higher secondary school	12.524082, 78.217945	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56140, 77.89991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59085, 77.90592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67883, 77.96865



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17041, 78.16823



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22515, 78.19158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22995, 78.19466



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24547, 78.19501



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24685, 78.19561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25120, 78.20010



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25442, 78.20015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32012, 78.20734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40661, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42548, 78.22666



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45066, 78.21821



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50265, 78.22345



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.50736, 78.22244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.50803, 78.22286



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51930, 78.22187



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.52231, 78.21904
